Gwynedd Mercy University-School of Business & Education
(*Must be used for each lesson observed by University Supervisor*)
Lesson Plan Framework

Subject Area: Learning Support Math **Grade Level:** 9th

Student Name: Grace Dittman

Domain I: Planning and Preparation	
PA Standard(s): ● **CC.2.2.HS.D.9:** Use reasoning to solve equations and justify the solution method. ● **CC.2.2.8.B.3:** Analyze and solve linear equations.	
Essential Question(s): Students will consider and reflect on throughout the lesson: ● How do inverse operations help me undo the steps in an equation to find the value of the variable? ● Why does the order of my steps matter when I solve a multistep equation? ● How can I check that my solution is correct?	
Objectives/Performance Expectations What will students know and be able to do as a result of this lesson? Students will be able to: ● Solve multistep equations by identifying and carrying out each step in the correct order using inverse operations. ● Explain or identify the next step in solving an equation. ● Check their answers and use correct solutions to complete the multistep equation color-by-number accurately.	
Assessment Evidence/Level of Learning Students will show what they know by evidence of/list names of formative or summative checks, quizzes, tests which indicate evidence of learning: ● **Formative:** Whiteboard responses during the warm-up: every student solves along with each step so I can see understanding at a glance. ● **Formative:** Questioning and observation in small groups (my notes and my cooperating teacher's notes).	**Differentiated Instruction**: Modifications in instruction and assessment for students with learning differences, specifically those with 504's and IEP's. Extensions/Enrichment in instruction and assessment for students who are above grade level and/or gifted. ● **Small groups:** Students are grouped by readiness so each group moves at the same pace. ● **Supports:** Visual schedule and a materials list are posted on the board.

● **Independent practice:** Multistep equation color-by-number: correct colors show whether each answer is right. ● **Practice data:** IXL recommendations for students who finish early. ● **Closing check:** Halloween Gimkit as a fun, low-pressure check of the skill.	● **Supports:** Each step is modeled aloud and students may use calculators. ● **Supports:** Students respond on whiteboards, which allows low-pressure practice and easy error correction. ● **Choice:** Students may choose their own seat as long as they are working. ● **IEP/504:** Accommodations in each student's IEP/504 plan will be followed (for example, extra time or chunked directions). ● **Extension:** Early finishers move on to their IXL recommendations.

Domain II: Classroom Environment:

Describe Classroom/Behavior Management Plan & Routines

- A visual schedule and a list of materials to grab are on the board when students enter, so routines are clear before class begins.
- Students gather whiteboards, markers, erasers, and calculators during the first 5 minutes to settle in.
- Students answer on whiteboards during the warm-up, and I invite volunteers to share the next step.
- Small groups are split by readiness, with one group working with me and the other with my cooperating teacher.
- After lunch, students choose their own seats as long as they are working. I circulate to help.
- Early finishers work on their IXL recommendations or on unfinished work.
- The class ends with a Halloween Gimkit celebration.

Domain III: Instruction

Activate Prior Knowledge/Motivation:

Preparing students for the lesson: How will you activate prior knowledge, build background, or review previous lessons as well as motivate students for the lesson?

The warm-up reviews the steps for solving multistep equations that students already know. Students answer on whiteboards and are invited to say the next step, so they stay actively involved. Working in small groups at the same pace lets each student build confidence. The color-by-number and the Halloween Gimkit make the practice fun and motivating.

Materials Needed:

Whiteboards, dry-erase markers, erasers, calculators, visual schedule and materials list (on the board), 4 warm-up multistep equations, multistep equation color-by-number, coloring supplies, pencils, student devices for IXL and Gimkit, Halloween Gimkit

Vocabulary (New or Review):

Equation; variable; constant; coefficient; inverse operation; isolate; solution; check

Sequence of the Lesson:
What learning activities/strategies will you use to engage the students in the learning? What will students do to use and apply new concepts or skills (independent practice if relevant)? How will you monitor and guide their performance? Include relevant vocabulary. Use bullets to sequence your lesson.

Arrival and Setup — 5 minutes

- Students read the visual schedule and materials list on the board.
- Students grab whiteboards, markers, erasers, and calculators and settle in.

Whole-Class Warm-Up — 10 minutes

- We solve 4 multistep equations together on the board.
- I go through each step aloud and give students a chance to tell me the next step or the answer.
- Students follow along on their whiteboards so I can see who understands and who needs support.

Small Group Instruction — 30 minutes

- The class splits into 2 small groups by readiness, so students in each group work at a similar pace.
- One group works with me and the other works with my cooperating teacher.
- Students practice multistep equations, explaining steps and checking their work with teacher support.

Lunch

Independent Practice: Color-by-Number — after lunch

- I hand out the multistep equation color-by-number.
- Students work individually and may choose their own seat as long as they are working.
- I circulate to help and check understanding.
- When finished, students work on IXL recommendations or any unfinished work.

Halloween Gimkit — end of class

- The class plays a fun Halloween Gimkit as a small celebration.

Closure: What will you do to bring closure to the lesson? How will you summarize this lesson and preview the lesson that will follow?

The class ends with a Halloween Gimkit that gives students one more chance to use their multistep equation skills while celebrating together. As students play, I can see which skills they know well and which need more practice.
Connection to Next Lesson: Teacher connects the lesson to continued practice solving multistep equations and checking solutions.

Reflection Guidelines

The student will write lesson plans with an eye to the Danielson Framework and the Four Domains of Professional Practice. Use this as a resource when writing lesson plans.

Domain I Planning and Preparation

1. How do you know you were knowledgeable of your content?
2. What is the evidence that your objective was clear in the lesson plan?
3. How does the lesson address individual student learning needs?
4. What variety of strategies and methodologies did you consider when planning?
5. What is the evidence that lesson components were logically sequenced?
6. How did your lesson plan include checks for student understanding?
7. What PA Standards did this lesson address?

Domain II Classroom Environment

1. How did students know the behavioral and academic expectations for the lesson?
2. What effective routines and procedures did you implement for lesson efficiency?
3. How did you insure the classroom was physically accessible for all learners?
4. What evidence indicates that respect was shown between and among students and teacher?
5. What evidence indicates that students placed a high value on quality participation, process, and product?
6. How do you assess if the lesson was attainable for all students but also had appropriate rigor?

Domain III Instructional Delivery

1. How did you align your lesson to PA Standards?
2. How did you insure a clearly stated objective, understood by learners?

3. How did you provide clear explanations as to how to proceed through the lesson?
4. What questioning and discussion techniques insured all-learner participation?
5. What strategies did you employ to engage all students?
6. Describe how you made the content meaningful due to sequencing and pacing.
7. Discuss how you checked for understanding, provided feedback, retaught.
8. What is the evidence that you were you flexible and responsive to the needs of the learners?
9. Describe how you made meaning of the lesson with some form of closure activity.

Overall Assessment

Briefly state any changes you will make the next time you teach this lesson or activity, to improve any of the Domains of Professional Practice.